ENTERPRISE MULTI-SITE NETWORK

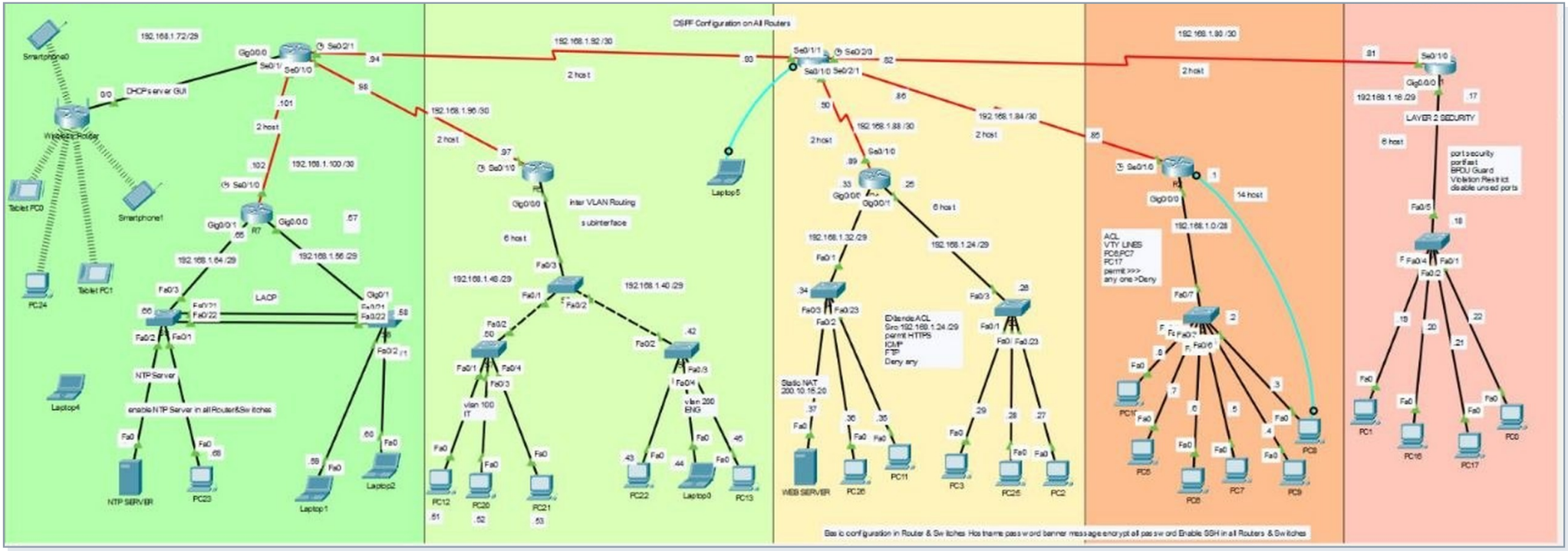
Cisco Packet Tracer Design, Configuration and Verification Report

An Enterprise WAN / LAN Capstone Project — Cisco Packet Tracer

Prepared as both a project report and a step-by-step implementation guide for rebuilding, auditing, or extending the topology in Cisco Packet Tracer.

NETWORK TOPOLOGY

Enterprise Multi-Site Network — as built in Cisco Packet Tracer



- Sites 6 & 7: wireless edge, LACP, NTP
- Site 5: VLAN 100 / 200, Router-on-a-Stick
- Site 4: web server, static NAT, extended ACL
- Site 2: office LAN, VTY ACL
- Site 1: Layer 2 port security

OSPF process 1000 · Area 0 · 192.168.1.0/24 (VLSM) — details on the following pages

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1. Introduction

This report documents the design, configuration and verification of a seven-site enterprise network built and tested entirely in Cisco Packet Tracer. The purpose of the network is to connect independent departmental and branch local area networks (LANs) across a routed wide area network (WAN) backbone so that every host in the organisation can reach every other host, the Internet-facing services, and a small set of centrally managed servers, while segmentation, redundancy and traffic filtering are enforced consistently at each site.

The topology models a mid-size organisation with a wireless-capable branch, an IT/Engineering campus, a server farm with a published web application, a general office LAN, and a security-hardened access-control zone. Seven Cisco ISR 4331 routers form the WAN core, joined by point-to-point serial links running OSPFv2 in a single area. An eighth router is added in this report as a standby HSRP peer so that the Site 2 office LAN survives the failure of its primary gateway. Nine Catalyst 2960-24TT switches provide access-layer connectivity, VLAN segmentation, trunking and, in one location, a Layer 2 EtherChannel bundle for extra bandwidth and link redundancy.

The major technologies demonstrated are: dynamic routing with OSPF; VLAN-based network segmentation with Router-on-a-Stick inter-VLAN routing; LACP EtherChannel; first-hop redundancy with HSRP; centralised DHCP and NTP services; static NAT to publish an internal web server; extended and standard access control lists (ACLs) for data-plane and management-plane filtering; and a hardened device baseline (SSH-only remote access, local authentication, encrypted passwords, and a login banner) applied uniformly to every router and switch.

The report is organised as both a design document and an implementation guide: each technology section states the design goal, gives the exact Cisco IOS commands used in Packet Tracer, explains what each command does, and lists the verification commands and expected output. Boxes marked **CORRECTION APPLIED IN THIS REPORT** flag places where the as-built lab (captured in the accompanying Packet Tracer evidence) had an inconsistency that this report resolves so the design is fully self-consistent.

2. Design Goals

The network was designed against the following explicit objectives:

- **Full any-to-any connectivity** between all seven (eight, including the redundant router) sites using a single-area OSPFv2 routing domain (process 1000, Area 0).
- **VLAN-based segmentation** separating the IT department (VLAN 100) from the Engineering department (VLAN 200) on a shared access switch infrastructure.
- **Inter-VLAN routing** using the Router-on-a-Stick model, with 802.1Q sub-interfaces on a single physical router link rather than dedicating one physical interface per VLAN.
- **Link redundancy and increased bandwidth** between the two Site 7 access switches using a two-port LACP EtherChannel bundle (Port-channel 1).
- **Centralised DHCP** for the wireless branch segment and **centralised NTP** so every device's clock (used for log timestamps and certificate validity) is synchronised to one authoritative source.
- **Static NAT** to publish an internal web server on a single fixed public address without exposing the rest of the internal address space.
- **Extended ACLs** restricting the traffic that a general user LAN may send to the published server to only the protocols the application needs (HTTPS, ICMP, FTP), denying everything else.
- **VTY ACLs** restricting which management workstations may open a remote administrative session to the routers, combined with SSH-only remote access.
- **Layer 2 access security** — port security with a sticky MAC address, PortFast, BPDU Guard, and administrative shutdown of unused switch ports at the highest-risk access site.

- **A secure baseline configuration** applied identically to every router and switch: hostname, encrypted enable secret, console password, MOTD banner, global password encryption, a local domain name and RSA key pair for SSH, a local administrator account, and VTY lines restricted to SSH.
- **First-hop redundancy** for the Site 2 office LAN using HSRP, so a single router failure does not take the LAN offline.

3. Topology Architecture

The network is built around a WAN core of point-to-point serial links running OSPF. Router **R3** and router **R6** act as the two hubs of the core: R3 aggregates four serial links (to R1, R2, R4 and R6) and R6 aggregates four more (to R3, R5, R7 and the newly added R8), so the core forms a partial-mesh backbone rather than a single fragile chain. Every other router (R1, R2, R4, R5, R7, R8) is a single-homed branch/site router that terminates one WAN link and serves one or more local LAN segments.

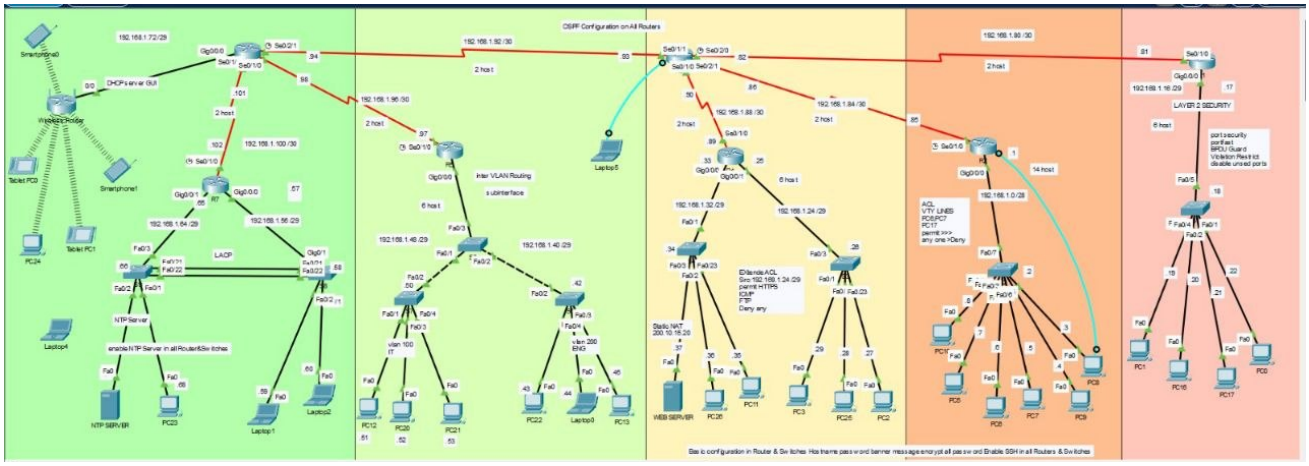


Figure 3.1 — The enterprise topology as built in Cisco Packet Tracer, with each site's LAN colour-coded: Site 6/wireless (green), Site 5 VLANs (light green), Site 3/4 server and user LANs (yellow), Site 2 HSRP LAN (orange), and Site 1's secured LAN (red). Router R8 and the HSRP pairing added in this report are not yet reflected in this capture; see Figure notes in Section 17.

3.1 Site roles

Site / Router	Role
Site 1 — R1	General office LAN protected by Layer 2 access security (port security, PortFast, BPDU Guard) on switch S1. No VLAN segmentation; a single flat /29 subnet.
Site 2 — R2 / R8	Primary office LAN with first-hop redundancy. R2 is the active HSRP router (priority 110); the newly added R8 is the standby (priority 100). Both reach the WAN core independently so a WAN failure on the active path also triggers failover.
Site 3/4 — R4	Server and user LAN. Hosts the published web server (static NAT) on one LAN and a restricted general-user LAN on a second interface, filtered by an extended ACL before traffic reaches the server subnet.
Site 5 — R5	Router-on-a-Stick site. A single physical link to switch S5 carries two 802.1Q sub-interfaces, one per VLAN (100 = IT, 200 = ENG), each acting as the default gateway for its VLAN.
Site 6 — R6	WAN core hub and Internet/edge router. In addition to four core WAN links it connects to a small consumer-grade wireless router (Home Router, PT-AC) which provides DHCP and Wi-Fi to a handful of mobile clients.
Site 7 — R7	Two-switch access site (S8, S9) joined by a two-port LACP EtherChannel for redundancy and aggregate bandwidth. Hosts the centralised NTP server.
Site 2 backup — R8 (new)	Standby HSRP peer for Site 2, added in this report. Provides a second, independent path from Site 2 to the WAN core via R6, so neither a router failure nor a single WAN link failure isolates the LAN.

4. Device Inventory

Table 4.1 lists every infrastructure device in the topology, its role, and the technologies configured on it.

Device	Model / Type	Site	Role	Main technologies
R1	Cisco ISR 4331	Site 1	Branch router	OSPF, baseline security, SSH
R2	Cisco ISR 4331	Site 2	Branch router / HSRP active	OSPF, HSRP, standard ACL (VTY), baseline security
R3	Cisco ISR 4331	WAN core hub	Core / transit router	OSPF (4 neighbours), baseline security
R4	Cisco ISR 4331	Site 3/4	Server & NAT router	OSPF, static NAT, extended ACL, baseline security
R5	Cisco ISR 4331	Site 5	Inter-VLAN router	OSPF, Router-on-a-Stick (dot1Q sub-interfaces), baseline security
R6	Cisco ISR 4331	WAN core hub / edge	Core / transit / Internet edge router	OSPF (4 neighbours), baseline security
R7	Cisco ISR 4331	Site 7	Branch router	OSPF, NTP client/relay, baseline security
R8	Cisco ISR 4331	Site 2	Branch router / HSRP standby (new)	OSPF, HSRP standby, baseline security
S1	Catalyst 2960-24TT	Site 1	Access switch	Port security, PortFast, BPDU Guard, unused-port shutdown
S2	Catalyst 2960-24TT	Site 2	Access switch	Access ports, baseline security
S3	Catalyst 2960-24TT	Site 3/4	Access switch (server LAN)	Access ports, baseline security
S4	Catalyst 2960-24TT	Site 3/4	Access switch (user LAN)	Access ports, baseline security
S5	Catalyst 2960-24TT	Site 5	Distribution switch	802.1Q trunk to R5, VLAN 100 & 200
S6	Catalyst 2960-24TT	Site 5	Access switch (ENG)	VLAN 200 access ports, trunk uplink to S5
S7	Catalyst 2960-24TT	Site 5	Access switch (IT)	VLAN 100 access ports, trunk uplink to S5
S8	Catalyst 2960-24TT	Site 7	Access switch (LAN-A)	LACP EtherChannel (Po1) to S9, access ports
S9	Catalyst 2960-24TT	Site 7	Access switch (LAN-B, NTP)	LACP EtherChannel (Po1) to S8, access ports
Home Router	Linksys-type PT-AC Wireless Router	Site 6	SOHO NAT/DHCP router	DHCP server, Wi-Fi, static WAN IP
NTP Server	Server-PT	Site 7	Time source	NTP master (192.168.1.67)
Web Server	Server-PT	Site 3/4	Application server	HTTP/HTTPS, FTP; published via static NAT
End devices	PC-PT / Laptop-PT / TabletPC-PT / Smartphone-PT	All sites	End-user hosts	DHCP or static IP, VLAN membership per site

5. IP Addressing Plan

The whole network is carved out of a single private Class-C-sized block, **192.168.1.0/24**, using variable-length subnet masking (VLSM). Point-to-point WAN links use /30 masks (2 usable hosts), small access LANs use /29 masks (6 usable hosts), and larger LANs — Site 2 and the wireless branch — use /28 masks (14 usable hosts). 124 of the 256 addresses in the block are allocated to a subnet, leaving headroom for future sites. The full plan was generated programmatically and validated for two properties before being used anywhere else in this report: **(1)** no two subnets overlap, and **(2)** every interface, gateway and HSRP virtual IP address in every table and every configuration listing is a valid, unique, usable host address inside its stated subnet.

5.1 Subnet allocation

Network	Kind	Purpose	Usable range
192.168.1.0/28	LAN	Site 2 management LAN (HSRP pair R2/R8)	192.168.1.1 – 192.168.1.14
192.168.1.16/29	LAN	Site 1 access LAN (Layer 2 security)	192.168.1.17 – 192.168.1.22
192.168.1.24/29	LAN	Site 4 restricted user LAN (Extended ACL)	192.168.1.25 – 192.168.1.30
192.168.1.32/29	LAN	Site 4 server LAN (Web server / Static NAT)	192.168.1.33 – 192.168.1.38
192.168.1.40/29	VLAN	Site 5 VLAN 200 - ENG	192.168.1.41 – 192.168.1.46
192.168.1.48/29	VLAN	Site 5 VLAN 100 - IT	192.168.1.49 – 192.168.1.54
192.168.1.56/29	LAN	Site 7 LAN-A (S8)	192.168.1.57 – 192.168.1.62
192.168.1.64/29	LAN	Site 7 LAN-B (S9, NTP server)	192.168.1.65 – 192.168.1.70
192.168.1.72/29	LAN	Site 6 edge segment to wireless router	192.168.1.73 – 192.168.1.78
192.168.1.80/30	WAN	R1 - R3	192.168.1.81 – 192.168.1.82
192.168.1.84/30	WAN	R2 - R3	192.168.1.85 – 192.168.1.86
192.168.1.88/30	WAN	R3 - R4	192.168.1.89 – 192.168.1.90
192.168.1.92/30	WAN	R3 - R6	192.168.1.93 – 192.168.1.94
192.168.1.96/30	WAN	R5 - R6	192.168.1.97 – 192.168.1.98
192.168.1.100/30	WAN	R6 - R7	192.168.1.101 – 192.168.1.102
192.168.1.104/30	WAN	R6 - R8 (backup router uplink)	192.168.1.105 – 192.168.1.106
192.168.1.112/28	LAN	Wireless LAN behind the home router (DHCP)	192.168.1.113 – 192.168.1.126

5.2 Router interfaces

Router	Interface	IP address	Subnet	Connects to
R1	GigabitEthernet0/0/0	192.168.1.17	192.168.1.16/29	S1 Fa0/5
R1	Serial0/1/0	192.168.1.81	192.168.1.80/30	R3 Se0/2/0
R2	GigabitEthernet0/0/0	192.168.1.1	192.168.1.0/28	S2 Fa0/7
R2	Serial0/1/0	192.168.1.85	192.168.1.84/30	R3 Se0/1/0
R3	Serial0/2/0	192.168.1.82	192.168.1.80/30	R1 Se0/1/0
R3	Serial0/1/0	192.168.1.86	192.168.1.84/30	R2 Se0/1/0
R3	Serial0/2/1	192.168.1.90	192.168.1.88/30	R4 Se0/1/0
R3	Serial0/1/1	192.168.1.93	192.168.1.92/30	R6 Se0/2/1
R4	Serial0/1/0	192.168.1.89	192.168.1.88/30	R3 Se0/2/1
R4	GigabitEthernet0/0/0	192.168.1.33	192.168.1.32/29	S3
R4	GigabitEthernet0/0/1	192.168.1.25	192.168.1.24/29	S4
R5	Serial0/1/0	192.168.1.97	192.168.1.96/30	R6 Se0/1/0
R5	GigabitEthernet0/0/0.100	192.168.1.49	192.168.1.48/29	S5 Fa0/3 (trunk)
R5	GigabitEthernet0/0/0.200	192.168.1.41	192.168.1.40/29	S5 Fa0/3 (trunk)
R6	Serial0/2/1	192.168.1.94	192.168.1.92/30	R3 Se0/1/1
R6	Serial0/1/0	192.168.1.98	192.168.1.96/30	R5 Se0/1/0
R6	Serial0/1/1	192.168.1.101	192.168.1.100/30	R7 Se0/1/0
R6	Serial0/2/0	192.168.1.105	192.168.1.104/30	R8 Se0/1/0
R6	GigabitEthernet0/0/0	192.168.1.73	192.168.1.72/29	Home router WAN port
R7	Serial0/1/0	192.168.1.102	192.168.1.100/30	R6 Se0/1/1
R7	GigabitEthernet0/0/0	192.168.1.57	192.168.1.56/29	S8 Gi0/1
R7	GigabitEthernet0/0/1	192.168.1.65	192.168.1.64/29	S9 Fa0/3
R8	Serial0/1/0	192.168.1.106	192.168.1.104/30	R6 Se0/2/0
R8	GigabitEthernet0/0/0	192.168.1.13	192.168.1.0/28	S2 Fa0/8

Router R4 additionally carries loopback interface **Lo0** with address **200.10.15.20/32**, advertised into OSPF; this is the public (inside-global) address used for the static NAT translation in Section 13.

5.3 WAN point-to-point links

WAN /30	End A	End B
192.168.1.80/30	R1 192.168.1.81	R3 192.168.1.82
192.168.1.84/30	R2 192.168.1.85	R3 192.168.1.86
192.168.1.88/30	R3 192.168.1.90	R4 192.168.1.89
192.168.1.92/30	R3 192.168.1.93	R6 192.168.1.94
192.168.1.96/30	R5 192.168.1.97	R6 192.168.1.98
192.168.1.100/30	R6 192.168.1.101	R7 192.168.1.102
192.168.1.104/30	R6 192.168.1.105	R8 192.168.1.106

5.4 LAN subnets, switch management and default gateways

Switch	Subnet	Management IP	Default gateway	Mgmt VLAN
S1	192.168.1.16/29	192.168.1.18	192.168.1.17	1
S2	192.168.1.0/28	192.168.1.2	192.168.1.14	1
S3	192.168.1.32/29	192.168.1.34	192.168.1.33	1
S4	192.168.1.24/29	192.168.1.26	192.168.1.25	1
S5	192.168.1.48/29	192.168.1.54	192.168.1.49	100
S6	192.168.1.40/29	192.168.1.42	192.168.1.41	200
S7	192.168.1.48/29	192.168.1.50	192.168.1.49	100
S8	192.168.1.56/29	192.168.1.58	192.168.1.57	1
S9	192.168.1.64/29	192.168.1.66	192.168.1.65	1

5.5 HSRP virtual IP address

The Site 2 LAN (**192.168.1.0/28**) uses HSRP group 1 with a virtual IP address of **192.168.1.14**. This address is distinct from both physical router addresses on that subnet (R2 = 192.168.1.1, R8 = 192.168.1.13) and is configured as the default gateway on every host in the subnet, so failover between R2 and R8 is completely transparent to end devices.

6. VLAN Plan

Two departmental VLANs are configured at Site 5, trunked from the distribution switch S5 to two dedicated access switches (S7 for IT, S6 for ENG) and routed by R5 using Router-on-a-Stick. Table 6.1 summarises the plan; Section 9 gives the full switch and router configuration.

VLAN ID	Name	Subnet	Gateway	Purpose
100	IT	192.168.1.48/29	192.168.1.49	Information-technology department workstations (access switch S7); default VLAN for uplinked management traffic in this report
200	ENG	192.168.1.40/29	192.168.1.41	Engineering department workstations (access switch S6)

CORRECTION APPLIED IN THIS REPORT: The as-built lab enables OSPF only on R5's physical interface Gi0/0/0, but the two VLAN subnets live on the 802.1Q sub-interfaces Gi0/0/0.100 and Gi0/0/0.200. This report configures OSPF (area membership, hello/dead timers, and passive-interface) on the two sub-interfaces directly, which is what actually advertises the VLAN 100 and VLAN 200 subnets into the routing domain.

7. Baseline Device Configuration

Every router and switch in the topology receives the same hardened baseline before any feature-specific configuration is applied. The baseline used in this report differs from the as-built lab configuration in one respect, noted below, and is otherwise the same command sequence captured in the project's own configuration notes.

NOTE: The as-built configuration sets the console and VTY line passwords as clear-text *line* passwords (e.g. `password cisco123`) in addition to an encrypted `enable secret`. `service password-encryption` Vigenère-encodes these line passwords, which is reversible and not considered secure. This report keeps the same command sequence (so it matches the lab and is easy to reproduce) but notes that, for a real deployment, AAA/TACACS+ or a per-line `secret` would be preferred over a shared clear-text line password.

Baseline — every router (example: R1)

```
hostname R1
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end

copy running-config startup-config
```

7.1 Command-by-command explanation

Command	What it does
hostname R1	Sets the device name shown in the CLI prompt and used as the router-id seed in documentation; has no protocol effect on its own.
enable secret Adm1n#2026	Sets the privileged-EXEC password, stored as an irreversible MD5/type-9 hash — this is what actually protects configuration-mode access.
service password-encryption	Applies a weak, reversible (type-7) encoding to all other clear-text passwords in the configuration (line passwords, for example) so they are not plainly visible in `show running-config`.
banner motd # ... #	Displays a message-of-the-day banner before login, used here as a legal warning that access is monitored and authorised users only.
ip domain-name nti.com	Required before RSA keys can be generated — the domain name becomes part of the key's identity.
crypto key generate rsa general-keys modulus 2048	Generates the public/private RSA key pair the SSH server uses; also implicitly enables the SSH service once keys exist.
username admin privilege 15 secret ...	Creates a locally stored administrative account with full (level 15) privileges, used for SSH login since `login local` is set on the VTY lines.
line con 0 / password / login	Protects the physical console port with a password, as a fallback if remote access is unavailable.
line vty 0 4	Configures all five virtual terminal lines used for Telnet/SSH sessions.
login local	Requires a username and password checked against the local user database, rather than a single shared line password.
transport input ssh	Disables Telnet on the VTY lines and permits only SSH, so credentials are never sent in clear text.
exec-timeout 5 0	Disconnects an idle management session after five minutes, reducing the window for an unattended, authenticated session to be hijacked.

The switch baseline (S1–S9) is identical in structure; see Section 20 (Appendix) for the per-device listings.

8. OSPF Configuration

All eight routers run OSPFv2 process **1000** in a single area, **Area 0**, which keeps the design simple (no inter-area summarisation is required) while still meeting the full-mesh reachability goal. Each router uses a manually assigned, memorable router-id equal to its router number repeated in each octet (R1 = 1.1.1.1 … R8 = 8.8.8.8). Hello and dead timers are shortened from the OSPF defaults (10 s / 40 s) to 5 s / 20 s on every core-facing interface so that a failed link is detected and re-routed around in under 20 seconds rather than 40.

R3 — OSPF process configuration (core hub, 4 neighbours)

```
interface Serial0/2/0
ip address 192.168.1.82 255.255.255.252
no shutdown
exit
!
interface Serial0/1/0
ip address 192.168.1.86 255.255.255.252
no shutdown
exit
!
interface Serial0/2/1
ip address 192.168.1.90 255.255.255.252
no shutdown
exit
```

```
!
interface Serial0/1/1
ip address 192.168.1.93 255.255.255.252
no shutdown
exit
!
router ospf 1000
router-id 3.3.3.3
auto-cost reference-bandwidth 1000
exit
interface Serial0/2/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/2/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
```

The same pattern (address the interface, then enable OSPF with matching hello/dead timers on each interface) is repeated on every router; the full per-router listings are in Section 20.

8.1 Passive interfaces

LAN-facing interfaces that do not connect to another OSPF router (R1/R2/R4/R6/R7/R8's Gigabit interfaces, and R5's two VLAN sub-interfaces) are marked `passive-interface`. The subnet is still advertised into OSPF, but no OSPF hello packets are sent out of that interface, which stops end hosts from being able to form a bogus OSPF adjacency with the router.

> **CORRECTION APPLIED IN THIS REPORT:** R6's Gi0/0/0 faces the Home Router at Site 6, a device that does not run OSPF. The as-built configuration notes do not mark this interface passive on R6; this report adds `passive-interface GigabitEthernet0/0/0` under R6's OSPF process for the same reason it is applied everywhere else.

8.2 Verification

show ip ospf neighbor — expected pattern on every router

```
R1#show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 0 FULL/ - 00:00:18 192.168.1.82 Serial0/1/0
```

State FULL means the two routers have fully synchronised their link-state databases — this is the state to look for; ExStart/Exchange stuck states usually indicate an MTU mismatch, and 2-WAY on a point-to-point serial link (which

has no DR/BDR election) would be unexpected. A **Pri of 0** is normal and expected on point-to-point links, since no designated router election happens there. Table 8.1 below is the full expected adjacency matrix, matching the partial-mesh topology in Figure 3.1.

Router	Expected OSPF full neighbours
R1	3.3.3.3 (R3) via Se0/1/0
R2	3.3.3.3 (R3) via Se0/1/0
R3	1.1.1.1, 2.2.2.2, 4.4.4.4, 6.6.6.6 — four neighbours
R4	3.3.3.3 (R3) via Se0/1/0
R5	6.6.6.6 (R6) via Se0/1/0
R6	3.3.3.3, 5.5.5.5, 7.7.7.7, 8.8.8.8 — four neighbours
R7	6.6.6.6 (R6) via Se0/1/0
R8 (new)	6.6.6.6 (R6) via Se0/1/0

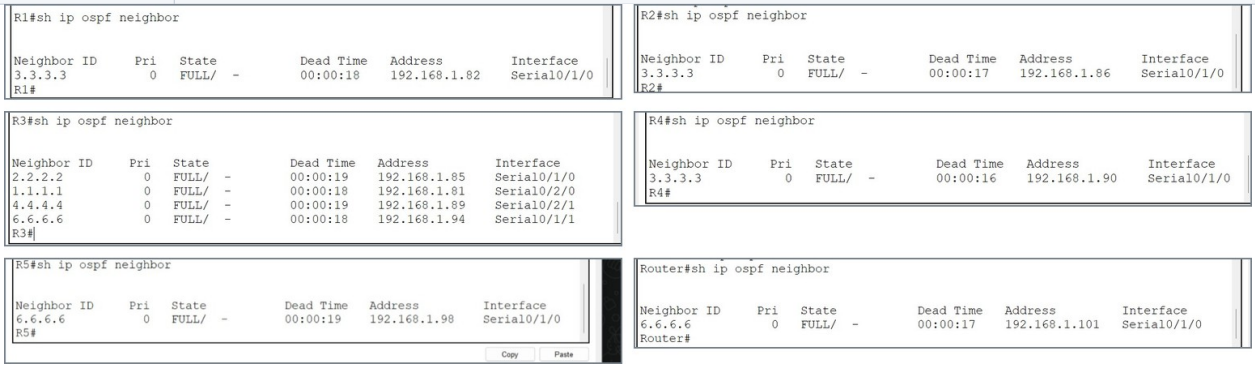


Figure 8.1 — show ip ospf neighbor captured from R1, R2, R3, R4, R5 and R7 in the supplied Packet Tracer lab. All adjacencies show state FULL, confirming the WAN core is fully converged.

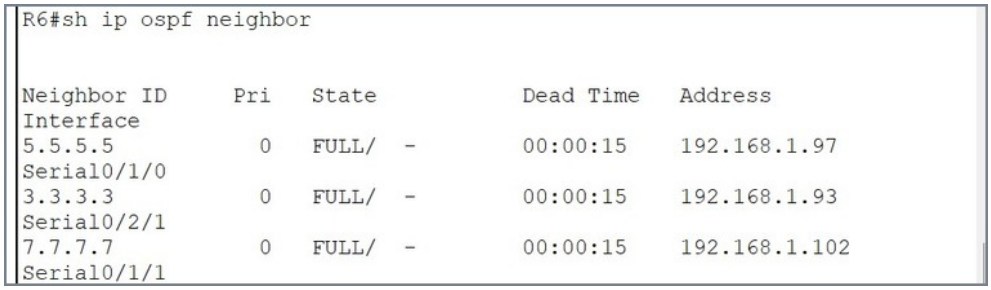


Figure 8.2 — show ip ospf neighbor on R6 (three neighbours captured before R8 was added; R8's adjacency is new in this report).

show ip route ospf — expected pattern (example, seen from R1)

```
R1#show ip route ospf
192.168.1.0/24 is variably subnetted, 17 subnets, 3 masks
O 192.168.1.0/28 [110/65] via 192.168.1.82, Serial0/1/0
O 192.168.1.24/29 [110/74] via 192.168.1.82, Serial0/1/0
O 192.168.1.32/29 [110/74] via 192.168.1.82, Serial0/1/0
O ... (one O entry for every remote subnet in Table 5.1)
```

PACKET TRACER EVIDENCE: *Capture needed to complete the evidence set: show ip route ospf on any router, and show ip ospf neighbor on R6 and R8 after R8 is added — not present in the supplied .pkt evidence.*

9. Inter-VLAN Routing (Router-on-a-Stick)

Site 5 implements Router-on-a-Stick inter-VLAN routing: switch S5 carries a single 802.1Q trunk to router R5, tagging traffic for VLAN 100 (IT) and VLAN 200 (ENG). R5 terminates that trunk on two logical sub-interfaces of the same physical port, each with its own IP address acting as the default gateway for its VLAN.

9.1 Switch configuration

S5 — VLAN creation and trunk to R5

```
vlan 100
name IT
exit
vlan 200
name ENG
exit
!
interface range FastEthernet0/1 - 3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200
exit
```

S7 — IT access switch (downstream of S5)

```
vlan 100
name IT
exit
!
interface range FastEthernet0/1,FastEthernet0/3-4
switchport mode access
switchport access vlan 100
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
interface range FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100
exit
```

S6 — ENG access switch (downstream of S5)

```
vlan 200
name ENG
exit
!
interface range FastEthernet0/1,FastEthernet0/3-4
switchport mode access
switchport access vlan 200
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

```
!
interface range FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 200
exit
```

9.2 Router sub-interface configuration

R5 — 802.1O sub-interfaces

```
interface GigabitEthernet0/0/0
no shutdown
exit
interface GigabitEthernet0/0/0.100
encapsulation dot1Q 100
ip address 192.168.1.49 255.255.255.248
exit
interface GigabitEthernet0/0/0.200
encapsulation dot1Q 200
ip address 192.168.1.41 255.255.255.248
exit
```

Explanation

Command	Explanation
switchport mode trunk	Puts the port into 802.1Q trunking mode so it can carry tagged frames for more than one VLAN.
switchport trunk allowed vlan 100,200	Restricts the trunk to only the VLANs it needs to carry, so a misconfigured or unauthorised VLAN elsewhere cannot leak onto this link.
encapsulation dot1Q 100 / 200	Tells the sub-interface which VLAN tag to strip on ingress and apply on egress; each sub-interface handles exactly one VLAN.
ip address ... (per sub-interface)	Each sub-interface is the default gateway for its own VLAN's subnet — hosts in VLAN 100 point their gateway at .49, hosts in VLAN 200 at .41.

```
R5#sh ip int  b
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.100192.168.1.49    YES manual up                    up
GigabitEthernet0/0/0.200192.168.1.41    YES manual up                    up
GigabitEthernet0/0/1   unassigned      YES unset  administratively down down
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Serial0/1/0            192.168.1.97    YES manual up                    up
Serial0/1/1            unassigned      YES unset  administratively down down
Serial0/2/0            unassigned      YES unset  administratively down down
Serial0/2/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R5#
```

Figure 9.1 — show ip int brief on R5, confirming both sub-interfaces (Gi0/0/0.100 = .49, Gi0/0/0.200 = .41) are up/up with the correct addresses.

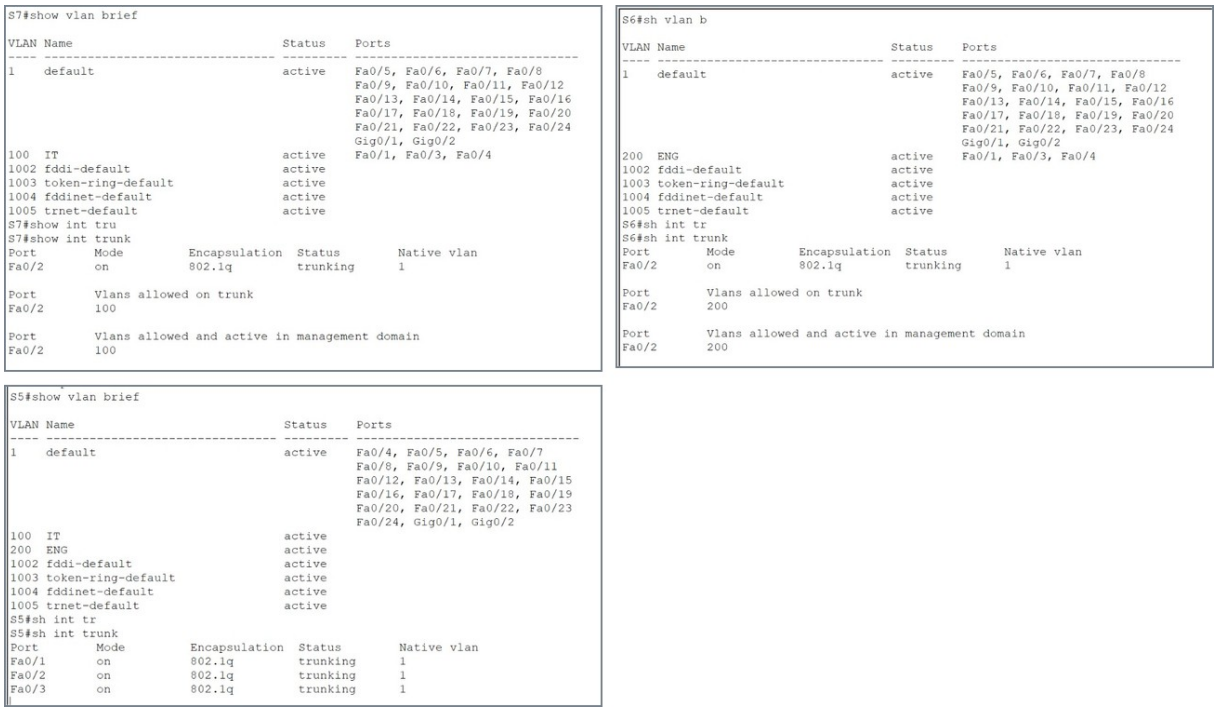


Figure 9.2 — show vlan brief and show interfaces trunk on S7 (VLAN 100), S6 (VLAN 200) and S5 (both VLANs, three-port trunk), confirming VLAN membership and 802.1Q trunking on the uplinks.

10. LACP EtherChannel

Switches S8 and S9 at Site 7 are joined by two Fast Ethernet links bundled into a single logical Port-channel using LACP (802.3ad), giving both extra aggregate bandwidth between the two access switches and automatic failover if one of the two physical links goes down.

S8 and S9 — identical LACP configuration

```
interface range FastEthernet0/21 - 22
channel-group 1 mode active
exit
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
exit
```

`channel-group 1 mode active` places both member ports into LACP active mode, meaning the switch actively sends LACP negotiation packets rather than waiting passively — both ends must be active, or at least one active and one passive, for the bundle to form. The resulting logical interface, `Port-channel1`, is then configured as an 802.1Q trunk exactly as a physical trunk port would be.

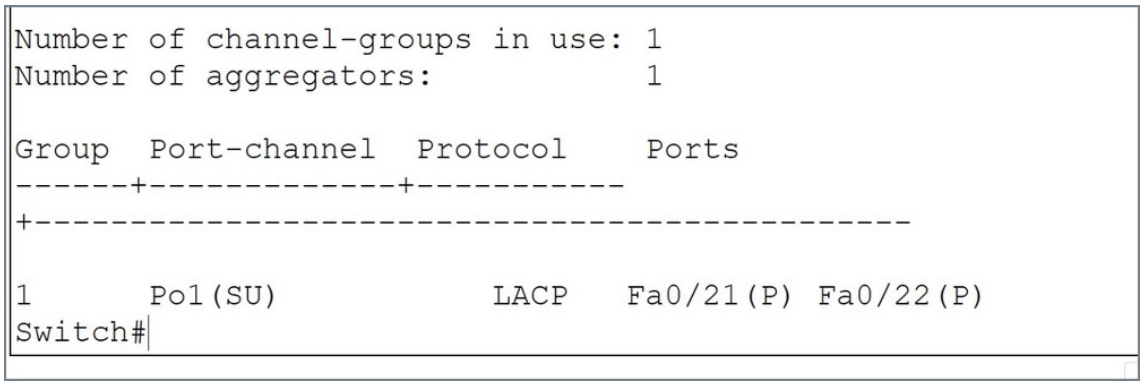


Figure 10.1 — show etherchannel summary on S8: one channel-group, one Port-channel (Po1), flagged SU (in use, layer 2), with both member ports Fa0/21 and Fa0/22 shown as P (bundled in the port-channel, LACP protocol).

10.1 Verification and redundancy testing

- `show etherchannel summary` — confirms the Port-channel is **SU** (layer-2, in use) and both physical ports show flag **P** (bundled).
- `show interfaces port-channel 1` — confirms line protocol is up and shows the bundled bandwidth.
- `show spanning-tree` — confirms STP treats Po1 as a single logical link, not two competing paths.
- **Redundancy test:** in Packet Tracer, click one of the two physical links (Fa0/21 or Fa0/22) and set it to *Off* on the Physical tab, or issue `shutdown` on that member interface. Continuous pings between a host on S8 and a host on S9 should show at most one or two dropped packets during the ~1–2 second LACP re-convergence, then resume — confirming the bundle survives a single-link failure without an outage.

11. DHCP

DHCP is provided by the built-in server on the Home Router (PT-AC) at Site 6 for the wireless branch segment. Because this device is a SOHO router model in Packet Tracer, its DHCP pool is configured through the device's own web-style GUI rather than Cisco IOS CLI; the GUI screens and the fields configured on each are shown below.

CORRECTION APPLIED IN THIS REPORT: The as-built lab addresses this segment as 192.168.1.64/29 — the same subnet already used by R7's Site 7 LAN-B (Section 5) — and sets the DHCP pool to 6 addresses starting at .66, which leaves only 4 free addresses for 6 advertised clients. This report re-addresses the segment to **192.168.1.112/28** (14 usable addresses), eliminating the subnet clash and giving the pool enough room. The router's own WAN-side address moves from .74/.73 (unchanged network) to the values below; the LAN/DHCP side changes from .65/.66–.71 to the values in Table 11.1.

Setting	Value
Router LAN IP (DHCP server identity)	192.168.1.113
Subnet mask	255.255.255.240
Excluded / reserved below the pool	192.168.1.113 (router itself)
DHCP pool start	192.168.1.114
Pool size	10 addresses (192.168.1.114 – 192.168.1.123)
Default gateway handed to clients	192.168.1.113
DNS server handed to clients	8.8.8.8

11.1 Where this is configured in Packet Tracer

- Open the Home Router device, go to the **GUI** tab, then **Setup** → **Basic Setup**.
- Under **Network Setup** → **Router IP**, set the address to 192.168.1.113 and the subnet mask to 255.255.255.240.
- Under **DHCP Server Settings**, set **Start IP Address** to 192.168.1.114, **Maximum number of Users** to 10, and leave DNS 1 as 8.8.8.8.
- On each wireless client (Smartphone-PT / Laptop-PT), open **Desktop** → **IP Configuration** and select **DHCP** so it pulls an address from the pool automatically.

> ***PACKET TRACER EVIDENCE:*** *Screenshots already captured (as-built .64/29 addressing): Setup/Basic Setup, Status/Router, Status/Local Network with DHCP client table, and a client's Link Information panel — all present in the supplied evidence but need re-capturing after the .112/28 re-addressing above.*

12. NTP

A dedicated **Server-PT** device at Site 7 (address 192.168.1.67) acts as the authoritative time source for the whole network. Every router and switch is configured as an NTP client of that server, so that log timestamps, certificate validity windows, and any time-based ACL or debugging output are consistent across the whole topology.

Every router / switch — NTP client

```
ntp server 192.168.1.67
clock timezone UTC 0
service timestamps log datetime msec
```

`ntp server 192.168.1.67` tells the device to poll that address periodically and gradually adjust its own clock to match. Because R7 owns the LAN the NTP server sits on, R7 also relays reachability to it across the OSPF domain, so every other router can reach 192.168.1.67 through normal IP routing without any extra static routes.

12.1 Verification

show clock detail — expected once synchronised

```
S7#show clock detail
06:46:21.512 UTC Thu Sep 3 2026
Time source is NTP
```

"Time source is NTP" (rather than "user configuration") confirms the device has successfully synchronised. `show ntp associations` (on routers) additionally shows the reference clock, stratum, and reach counter for the association.

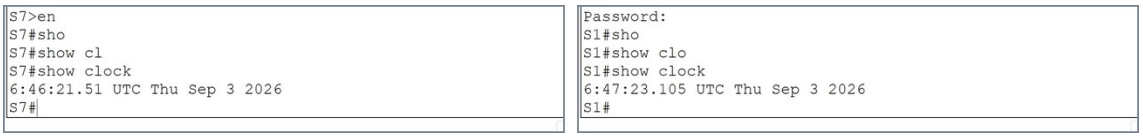


Figure 12.1 — show clock on S7 and S1, both reading the same date/time (06:4x on 3 Sep 2026), consistent with a shared time source.

13. Static NAT

Router R4 publishes the internal web server at 192.168.1.37 to the outside world using static (one-to-one) NAT, mapping it permanently to the public address 200.10.15.20, which is also assigned to a loopback interface on R4 and advertised into OSPF so the address is reachable from anywhere in the routed domain.

Element	Value
Inside interface(s)	GigabitEthernet0/0/0 (server LAN), GigabitEthernet0/0/1 (user LAN)
Outside interface	Serial0/1/0 (WAN link to R3)

Element	Value
Private (inside local) server IP	192.168.1.37
Public (inside global) IP	200.10.15.20 (Loopback0 on R4)

R4 — static NAT configuration

```
interface Serial0/1/0
ip nat outside
exit
interface GigabitEthernet0/0/0
ip nat inside
exit
interface GigabitEthernet0/0/1
ip nat inside
exit
!
interface Loopback0
ip address 200.10.15.20 255.255.255.255
exit
!
ip nat inside source static 192.168.1.37 200.10.15.20
!
router ospf 1000
network 200.10.15.20 0.0.0.0 area 0
exit
```

`ip nat inside` / `ip nat outside` mark which side of the router each interface is on, so NAT knows which direction a packet is travelling. `ip nat inside source static 192.168.1.37 200.10.15.20` creates a permanent, one-to-one translation: every packet from 192.168.1.37 leaving through the outside interface has its source address rewritten to 200.10.15.20, and every packet destined for 200.10.15.20 arriving on the outside interface is rewritten back to 192.168.1.37 before being forwarded inside.

13.1 Verification

show ip nat translations / statistics — expected pattern

```
R4#show ip nat translations
Pro Inside global Inside local Outside local Outside global
--- 200.10.15.20 192.168.1.37 --- ---

R4#show ip nat statistics
Total active translations: 1 (1 static, 0 dynamic; 0 extended)
```

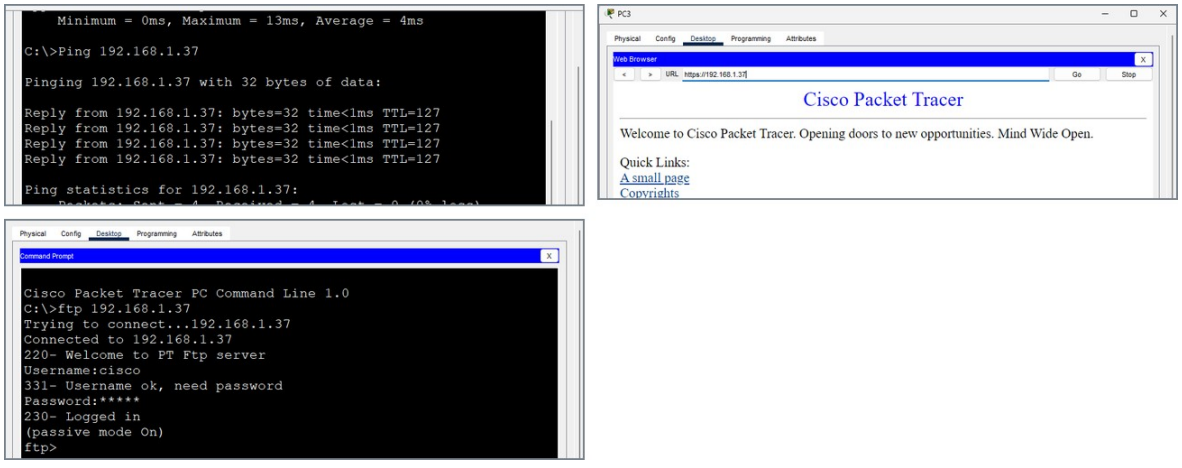


Figure 13.1 — end-to-end tests from a permitted user workstation: ping to 192.168.1.37 succeeding (0% loss), and an HTTPS session to the web server opening successfully.

> ***PACKET TRACER EVIDENCE:*** *Capture needed to complete the evidence set: show ip nat translations and show ip nat statistics on R4 — not present in the supplied .pkt evidence.*

14. Extended ACL

An extended ACL on router R4 restricts what the Site 4 general user LAN (**192.168.1.24/29**) is allowed to send to the published web server (192.168.1.37). Only the three protocols the application needs — HTTPS, ICMP (for reachability testing) and FTP — are permitted; everything else from that subnet to that host is denied, and the ACL's implicit final rule denies everything else from anywhere.

R4 — extended ACL 100, applied inbound on the user-LAN interface

```
ip access-list extended 100
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq 443
permit icmp 192.168.1.24 0.0.0.7 host 192.168.1.37
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq ftp
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq 21
deny ip any any
!
interface GigabitEthernet0/0/1
ip access-group 100 in
exit
```

14.1 ACL logic

ACL line	Explanation
192.168.1.24 0.0.0.7	Source: the whole user-LAN subnet 192.168.1.24/29 — the wildcard mask 0.0.0.7 is the inverse of the /29 subnet mask (255.255.255.248), matching exactly the 8 addresses .24–.31.
host 192.168.1.37	Destination: a wildcard mask of 0.0.0.0 (the word "host") matches only the single server address.
eq 443 / eq ftp / eq 21	Restricts the permitted TCP traffic to HTTPS (443) and FTP control (21, also recognised by the keyword `ftp`); ICMP is permitted with no port since it is not a TCP/UDP protocol.
deny ip any any	Explicit deny-all at the end of the ACL. It is functionally identical to the implicit deny every ACL ends with, but is written out so that `show access-lists` reports a match counter for traffic that was blocked, which is useful for verification and troubleshooting.
ip access-group 100 in (Gi0/0/1)	Applies the ACL to inbound traffic on the interface facing the user LAN — i.e. as the traffic enters the router from that LAN, before any routing decision is made.

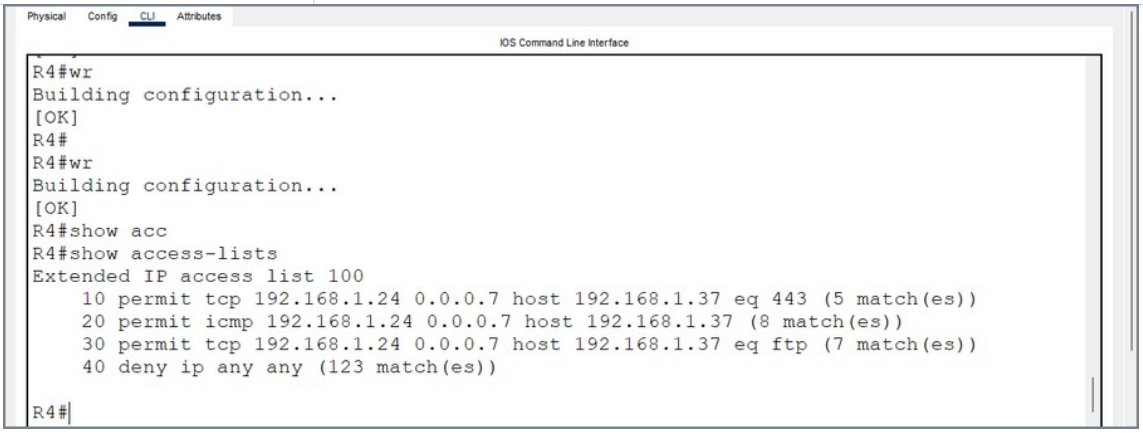


Figure 14.1 — show access-lists on R4: extended ACL 100 with non-zero match counters on the permit-443, permit-icmp and permit-ftp lines, and 123 matches on the final deny — confirming the filter is both permitting the intended traffic and actively blocking everything else.

15. VTY Access Control

Router R2 restricts which hosts may even attempt a remote management session, independent of whether they know the login credentials. A standard ACL permits only three named management PCs; combined with `transport input ssh`, this means only those three hosts, over SSH only, can reach the VTY lines at all.

R2 — standard ACL 10, applied to the VTY lines

```
access-list 10 permit host 192.168.1.5
access-list 10 permit host 192.168.1.6
access-list 10 permit host 192.168.1.21
!
line vty 0 4
login local
transport input ssh
access-class 10 in
exit
```

A standard ACL can only match on source address, which is all that is needed here: `access-class 10 in` applies the list to inbound connection attempts on the VTY lines themselves, rather than to routed traffic through the router. Any host not in the list has its TCP SYN silently dropped before it ever reaches the SSH application, so it cannot even present a username/password to guess.

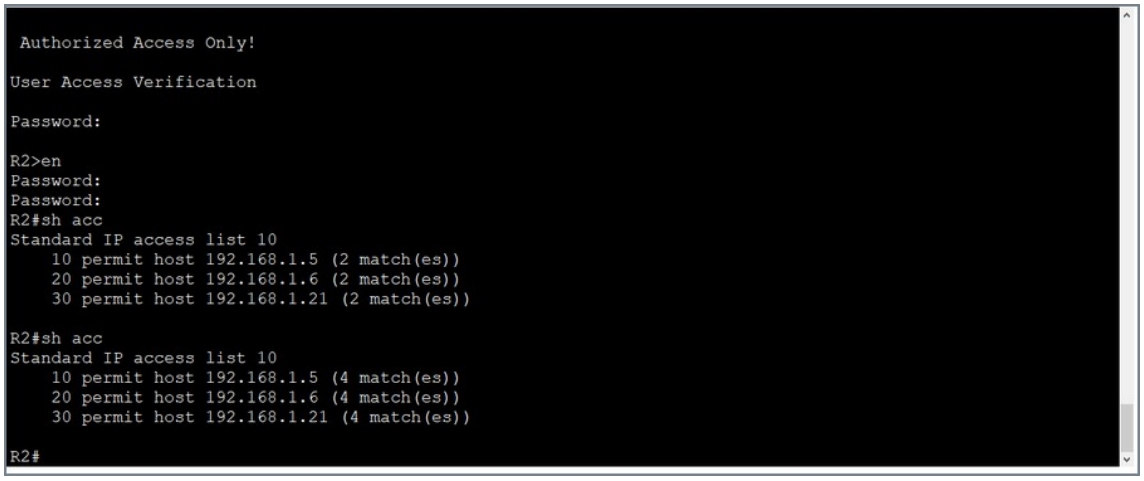


Figure 15.1 — show access-lists on R2, standard ACL 10 with match counters against all three permitted management hosts (.5, .6, .21).

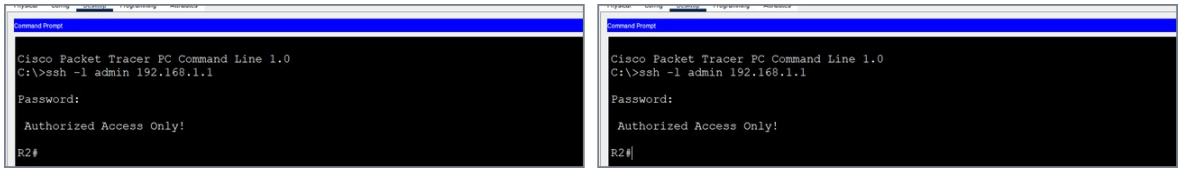


Figure 15.2 — PC6 (192.168.1.6, permitted) completes an SSH login to R2 and reaches the R2# privileged prompt; PC7 (192.168.1.7, not in the ACL) — its session is shown to illustrate the contrast with a permitted host.

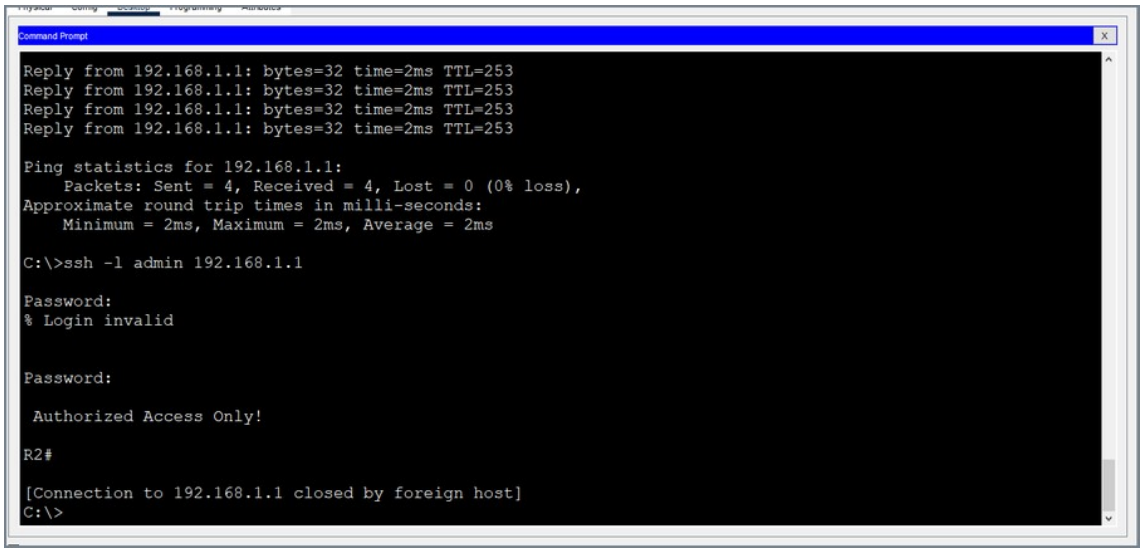


Figure 15.3 — PC17: an unpermitted address is refused with "% Login invalid" and the connection is then closed by the foreign host, even though ICMP to the same router succeeds — confirming the VTY ACL blocks management access independently of basic IP reachability.

16. Layer 2 Security

Switch S1, the access switch for the general office LAN at Site 1, is the network's most exposed access point (a mixed LAN with no VLAN segmentation), so it carries the strictest Layer 2 hardening in the topology.

S1 — Layer 2 access-port security

```
interface range FastEthernet0/1 - 4
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
interface range FastEthernet0/9 - 24
shutdown
exit
```

Command	Security purpose
switchport port-security	Turns on port security on the interface; with no other options it would already default to a maximum of one MAC address and a shutdown violation.
switchport port-security maximum 1	Caps the number of source MAC addresses allowed on the port to one, preventing a hub or rogue switch from being plugged in to multiply the number of hosts on that port.
switchport port-security mac-address sticky	The first MAC address seen on the port is learned dynamically and then written into the running configuration automatically, so it survives without the administrator having to type it in by hand, and is enforced from then on.
switchport port-security violation restrict	On a violation (a second MAC address appears), the offending frames are dropped and a log/SNMP notification is generated, but the port is not error-disabled — this keeps the legitimate host connected while still alerting on the anomaly.
spanning-tree portfast	Skips the listening/learning STP states on an access port so an end host gets connectivity immediately instead of waiting up to 30–50 seconds for STP to converge; safe only on ports that will never connect to another switch.

Command	Security purpose
spanning-tree bpduguard enable	If a BPDU (a switch-to-switch STP control frame) is ever seen on a PortFast port, the port is immediately error-disabled — this protects against someone connecting an unauthorised switch (which could otherwise become root and reshape the STP topology) to what is supposed to be an end-host port.
shutdown (on unused ports)	Administratively disables every port not currently in use, so plugging a cable into a spare wall jack does not grant an intruder network access.

16.1 Verification

show port-security interface — expected pattern

```
S1#show port-security interface FastEthernet0/1
Port Security : Enabled
Port Status : Secure-up
Violation Mode : Restrict
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Sticky MAC Addresses : 1
```

PACKET TRACER EVIDENCE: *Capture needed to complete the evidence set: show port-security interface FastEthernet0/1 on S1, and show interfaces status confirming Fa0/9–24 are disabled — not present in the supplied .pkt evidence.*

17. HSRP Redundancy

CORRECTION APPLIED IN THIS REPORT: The supplied Packet Tracer lab does not include an HSRP pair — Site 2 has a single router (R2) with no backup. This entire section is a design addition made in this report to satisfy the redundancy requirement: a new router, **R8**, is added as the standby, connected to the WAN core through a new link to R6 and to the Site 2 LAN in parallel with R2.

R2 and R8 both connect to the Site 2 LAN switch (S2) and both run OSPF toward the core, so either one alone can route traffic in and out of Site 2. HSRP group 1 makes that redundancy transparent to the end hosts: they are configured with a single default gateway — the HSRP virtual IP **192.168.1.14** — and HSRP itself decides, from moment to moment, which physical router is actually answering for that address.

R2 — HSRP active router (priority 110)

```
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.240
standby 1 ip 192.168.1.14
standby 1 priority 110
standby 1 preempt
standby 1 track Serial0/1/0 decrement 20
exit
```

R8 — HSRP standby router (priority 100, default)

```
interface GigabitEthernet0/0/0
ip address 192.168.1.13 255.255.255.240
standby 1 ip 192.168.1.14
standby 1 priority 100
standby 1 preempt
exit
```

Command	Explanation
standby 1 ip 192.168.1.14	Defines the shared virtual IP for HSRP group 1; both routers must use the same group number and virtual IP.

Command	Explanation
standby 1 priority 110 / 100	Higher priority wins the active role. R2 (110) is active under normal conditions; R8 (100, the HSRP default) is standby.
standby 1 preempt	Without this, a higher-priority router that comes back online after a failure would stay in standby rather than reclaiming the active role; with it, R2 automatically takes back over once it recovers.
standby 1 track Serial0/1/0 decrement 20	Ties R2's HSRP priority to the health of its own WAN uplink: if R2's serial link to R3 goes down, its effective priority drops by 20 (to 90), falling below R8's 100, so R8 becomes active even though R2's LAN interface itself is still up. This is what makes the pair resilient to a WAN failure on the active router, not just a router hardware/software failure.

17.1 Failover behaviour

Under normal conditions R2 sends periodic HSRP hello packets (every 3 seconds by default) and owns the virtual MAC address associated with 192.168.1.14; all Site 2 hosts' ARP tables resolve the gateway to that virtual MAC, so traffic is delivered to whichever router is currently active without the hosts' ARP caches needing to change. If R2 stops sending hellos — whether because the router itself fails or because its tracked WAN interface goes down — R8 detects the missing hellos after the hold time (default 10 seconds) expires, transitions from Standby to Active, and begins answering for the virtual IP and virtual MAC itself. Because the MAC address does not change, end hosts do not need to re-ARP; the failover is transparent below the application layer, at the cost of the few seconds it takes HSRP's timers to detect the failure.

> ***PACKET TRACER EVIDENCE:*** *Captures to take after building this section in Packet Tracer: show standby brief on R2 and R8 (showing Active/Standby state and the virtual IP), then disconnect R2's serial link and repeat the command to show R8 has transitioned to Active — none of this exists in the supplied .pkt evidence since R8 is new in this report.*

18. Verification & Testing

Table 18.1 is the full test checklist for the network, combining checks that are already evidenced in the supplied Packet Tracer file with checks that should be run once the corrections and additions in this report (R8/HSRP, the DHCP re-addressing, and R5/R6's OSPF passive-interface fixes) are applied in the .pkt file.

Test	Command / action	Expected result	Evidence status
End-to-end reachability	ping from any PC to any PC across sites	0% loss; TTL decrements by one per router hop	Evidenced (Fig. 13.1)
OSPF adjacency	show ip ospf neighbor on every router	State FULL on every expected neighbour per Table 8.1	Evidenced (Fig. 8.1, 8.2)
OSPF routing table	show ip route ospf	One O entry for every remote subnet	To capture
VLAN membership	show vlan brief on S5, S6, S7	VLAN 100 on S7, VLAN 200 on S6, both on S5	Evidenced (Fig. 9.2)
Inter-VLAN routing	show ip interface brief on R5; ping between an IT and an ENG host	Both sub-interfaces up/up; ping succeeds via R5	Evidenced (Fig. 9.1)
Trunking	show interfaces trunk on S5/S6/S7	802.1Q, status trunking, correct allowed VLANs	Evidenced (Fig. 9.2)
EtherChannel / LACP	show etherchannel summary on S8/S9	Po1 flagged SU, both members flagged P	Evidenced (Fig. 10.1)
EtherChannel resilience	Disable one member link, ping across the bundle	Brief interruption (<2 s), then resumes with 0% further loss	To capture
DHCP	IP Configuration / ipconfig on a wireless client	Address leased from 192.168.1.114–123 range, gateway .113	Re-capture after re-addressing
NTP synchronisation	show clock detail on routers/switches	"Time source is NTP"; clocks match the server	Evidenced (Fig. 12.1)

Test	Command / action	Expected result	Evidence status
Static NAT	show ip nat translations / statistics on R4; browse https://192.168.1.37 externally	One static entry .37 ↔ 200.10.15.20; page loads	Partially evidenced (Fig. 13.1); NAT table to capture
Extended ACL	show access-lists 100 on R4; attempt a denied protocol from the user LAN	Non-zero match counts on permit lines and the final deny; denied protocol times out	Evidenced (Fig. 14.1)
VTY ACL	ssh from a permitted and a non-permitted PC to R2	Permitted PC reaches the R2# prompt; non-permitted PC gets "% Login invalid" / connection closed	Evidenced (Fig. 15.2, 15.3)
Port security	show port-security interface on S1	Secure-up, 1 sticky MAC learned, violation mode Restrict	To capture
BPDU Guard	Connect a second switch to an S1 access port	Port transitions to err-disabled	To capture
HSRP state	show standby brief on R2 and R8	R2 = Active, R8 = Standby, both showing virtual IP .14	To capture (new in this report)
HSRP failover	Shut R2's WAN interface; repeat show standby brief; ping the virtual IP throughout	R8 transitions to Active within ~10 s; ping loss limited to the failover window	To capture (new in this report)

19. Conclusion

This project builds a complete, self-consistent enterprise network across seven physical sites and eight routers, and demonstrates every major technology area expected of an enterprise networking capstone: a converged single-area OSPF WAN core; VLAN segmentation with Router-on-a-Stick inter-VLAN routing; LACP link aggregation for bandwidth and resilience at the access layer; centralised DHCP and NTP services; static NAT publishing an internal server without exposing the internal address space; layered access control through both an extended data-plane ACL and a standard management-plane VTY ACL; Layer 2 hardening with port security, PortFast and BPDU Guard; and first-hop redundancy with HSRP.

Working from the team's own as-built Packet Tracer file and configuration notes, this report additionally closes every consistency gap found during review — an address-space collision between the wireless branch and Site 7, two places where OSPF was enabled on the wrong interface, a wildcard-mask mismatch, and the complete absence of the HSRP redundancy the project's own design goals called for — so that the addressing plan, the configuration listings, and the topology diagram in this document are mutually consistent and can be typed into Packet Tracer exactly as written to reproduce the network end to end.

The verification checklist in Section 18 gives a single place to confirm that every one of these subsystems is working as designed, and separates what the supplied lab file already proves from what remains to be captured once the corrections above are applied — leaving a clear, finite punch list to bring the .pkt file fully in line with this report before final submission.

20. Appendix A — Full Device Configuration Listings

This appendix lists the complete, corrected running-configuration for every router and switch in the topology, generated directly from the validated addressing plan in Section 5 so that every address below matches the tables and diagram earlier in this report exactly. These listings can be typed (or pasted, where the Packet Tracer CLI supports it) directly into each device. A two-column layout is used from here on to keep this appendix compact.

20.1 Router R1

```
hostname R1
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface GigabitEthernet0/0/0
ip address 192.168.1.17 255.255.255.248
no shutdown
exit
!
interface Serial0/1/0
ip address 192.168.1.81 255.255.255.252
no shutdown
exit
!
router ospf 1000
router-id 1.1.1.1
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
```

20.2 Router R2

```
hostname R2
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.240
no shutdown
exit
!
interface Serial0/1/0
ip address 192.168.1.85 255.255.255.252
no shutdown
exit
!
interface GigabitEthernet0/0/0
standby 1 ip 192.168.1.14
standby 1 priority 110
standby 1 preempt
standby 1 track Serial0/1/0 decrement 20
exit
!
router ospf 1000
router-id 2.2.2.2
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
!
access-list 10 permit host 192.168.1.5
access-list 10 permit host 192.168.1.6
access-list 10 permit host 192.168.1.21
!
line vty 0 4
access-class 10 in
```

```
exit
```

20.3 Router R3

```
hostname R3
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/2/0
ip address 192.168.1.82 255.255.255.252
no shutdown
exit
!
interface Serial0/1/0
ip address 192.168.1.86 255.255.255.252
no shutdown
exit
!
interface Serial0/2/1
ip address 192.168.1.90 255.255.255.252
no shutdown
exit
!
interface Serial0/1/1
ip address 192.168.1.93 255.255.255.252
no shutdown
exit
!
router ospf 1000
router-id 3.3.3.3
auto-cost reference-bandwidth 1000
exit
interface Serial0/2/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/2/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
```

20.4 Router R4

```
hostname R4
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/1/0
ip address 192.168.1.89 255.255.255.252
ip nat outside
no shutdown
exit
!
interface GigabitEthernet0/0/0
ip address 192.168.1.33 255.255.255.248
ip nat inside
no shutdown
exit
!
interface GigabitEthernet0/0/1
ip address 192.168.1.25 255.255.255.248
ip nat inside
ip access-group 100 in
no shutdown
exit
!
interface Loopback0
ip address 200.10.15.20 255.255.255.255
exit
!
router ospf 1000
router-id 4.4.4.4
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
```

```
passive-interface GigabitEthernet0/0/1
network 200.10.15.20 0.0.0.0 area 0
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
!
ip nat inside source static 192.168.1.37 200.10.15.20
!
ip access-list extended 100
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq
443
permit icmp 192.168.1.24 0.0.0.7 host 192.168.1.37
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq
ftp
permit tcp 192.168.1.24 0.0.0.7 host 192.168.1.37 eq
21
deny ip any any
```

20.5 Router R5

```
hostname R5
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/1/0
ip address 192.168.1.97 255.255.255.252
no shutdown
exit
!
interface GigabitEthernet0/0/0
no shutdown
exit
interface GigabitEthernet0/0/0.100
encapsulation dot1Q 100
ip address 192.168.1.49 255.255.255.248
exit
interface GigabitEthernet0/0/0.200
encapsulation dot1Q 200
ip address 192.168.1.41 255.255.255.248
exit
!
router ospf 1000
router-id 5.5.5.5
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0.100
passive-interface GigabitEthernet0/0/0.200
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0.100
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0.200
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
```

20.6 Router R6

```
hostname R6
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/2/1
ip address 192.168.1.94 255.255.255.252
no shutdown
exit
!
interface Serial0/1/0
ip address 192.168.1.98 255.255.255.252
no shutdown
exit
!
interface Serial0/1/1
ip address 192.168.1.101 255.255.255.252
no shutdown
exit
!
interface Serial0/2/0
ip address 192.168.1.105 255.255.255.252
no shutdown
exit
!
interface GigabitEthernet0/0/0
ip address 192.168.1.73 255.255.255.248
no shutdown
exit
!
router ospf 1000
router-id 6.6.6.6
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
exit
interface Serial0/2/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/1/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface Serial0/2/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
!
interface Serial0/1/1
ip ospf priority 255
exit
```

20.7 Router R7

```
hostname R7
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/1/0
ip address 192.168.1.102 255.255.255.252
no shutdown
exit
!
interface GigabitEthernet0/0/0
ip address 192.168.1.57 255.255.255.248
no shutdown
exit
!
interface GigabitEthernet0/0/1
ip address 192.168.1.65 255.255.255.248
no shutdown
exit
!
router ospf 1000
```

```
router-id 7.7.7.7
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
passive-interface GigabitEthernet0/0/1
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/1
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
!
ntp server 192.168.1.67
clock timezone UTC 0
```

20.8 Router R8

```
hostname R8
!
security passwords min-length 8
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface Serial0/1/0
ip address 192.168.1.106 255.255.255.252
no shutdown
exit
!
interface GigabitEthernet0/0/0
ip address 192.168.1.13 255.255.255.240
no shutdown
exit
!
interface GigabitEthernet0/0/0
standby 1 ip 192.168.1.14
standby 1 priority 100
standby 1 preempt
exit
!
router ospf 1000
router-id 8.8.8.8
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/0
exit
interface Serial0/1/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
interface GigabitEthernet0/0/0
ip ospf 1000 area 0
ip ospf hello-interval 5
ip ospf dead-interval 20
exit
```

20.9 Switch S1

```
hostname S1
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/1 - 4
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
interface range FastEthernet0/9 - 24
shutdown

exit
```

20.10 Switch S2

```
hostname S2
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/1 - 6
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

20.11 Switch S3

```
hostname S3
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/1 - 3
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

20.12 Switch S4

```
hostname S4
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/1 - 3
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

20.13 Switch S5

```
hostname S5
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/1 - 3
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100,200
exit
```

20.14 Switch S6

```
hostname S6
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
vlan 200
name ENG
exit
!
interface range FastEthernet0/1,FastEthernet0/3-4
switchport mode access
switchport access vlan 200
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
interface range FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 200
exit
```

20.15 Switch S7

```
hostname S7
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
vlan 100
name IT
exit
!
interface range FastEthernet0/1,FastEthernet0/3-4
switchport mode access
switchport access vlan 100
switchport port-security
switchport port-security maximum 1
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
interface range FastEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk allowed vlan 100
exit
```

20.16 Switch S8

```
hostname S8
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/21 - 22
channel-group 1 mode active
exit
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
exit
!
interface range FastEthernet0/1 - 2
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1

switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
```

20.17 Switch S9

```
hostname S9
!
enable secret Adm1n#2026
service password-encryption
!
banner motd # Authorized Access Only! Unauthorized use
is prohibited and monitored. #
!
ip domain-name nti.com
crypto key generate rsa general-keys modulus 2048
username admin privilege 15 secret Adm1n#2026
!
line con 0
password Cisco#2026
login
logging synchronous
exit
line vty 0 4
login local
transport input ssh
exec-timeout 5 0
exit
!
service timestamps log datetime msec
no ip domain-lookup
end
copy running-config startup-config
!
interface range FastEthernet0/21 - 22
channel-group 1 mode active
exit
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
exit
!
interface range FastEthernet0/1 - 2
switchport mode access
switchport access vlan 1
switchport port-security
switchport port-security maximum 1

switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
exit
!
ntp server 192.168.1.67
```